

Fall Protection Plan Developer INDEX

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P1. A Brief History

Gravity Training was established in 1999 and is an IWH, Services SETA as well as MER Seta accredited Training Provider. In 2008 Gravity Training successfully complied with the South African Qualifications Authority Act (Act 58 of 1995) to provide quality outcomes-based training and assessment in accordance with the relevant Unit Standards as set forth by SETA.

Based on extensive experience gained, our accredited work at height courses including fall arrest (tower climbing), rope access, rope rigging, climbing equipment inspection and other courses are presented across South Africa, the African Continent and the International market.

Since the 1970's, companies realised that erecting scaffolding or hiring cranes in order to perform work at height was a very expensive and time consuming solution. Thus, the world of Work at Height was born. Since then the need for quality training in these fields steadily rose and is still doing so today.

This course, **Fall Protection Plan Developer**, will be focussing on the knowledge, legal requirements and regulations that govern work at height. As well as how to interpret, implement and write a Fall Protection Plan in order to establish and maintain a safe work environment.

P2. To The Learner

There are inherent risks involved in working at height and these risks are also ever -present, if not more so, during Training. To minimise these risks, experience and knowledge must be gained in order to safely navigate the hazards as they present themselves. In order to gain the required knowledge this manual has been created as an aid, **and only an aid for trained and experienced** work at height technicians and facilitators. It is not designed to be used in isolation or without the input from a person trained in using it. However, at Gravity Training, all of the facilitators undergo rigorous training programmes in order to ensure that they are well equipped and informed as possible to present this text in such a way that it is easy to understand, remember and implement on site. To ensure that this is indeed the outcome of this course, the onus lies upon the learner to dedicate him/herself for the duration of the course to understand and practise the techniques that will be demonstrated during the entirety of this course.

P3. Abbreviations and definitions

1.1. Abbreviations or Acronyms

- 1.1.1. **CMA** - Conveyer Manufacturing Association.
- 1.1.2. **COF** - Certificate of fitness
- 1.1.3. **CR** - Construction Regulations 2014 of the OHS act 85 of 1993
- 1.1.4. **DSTI** - Daily Safe Task Instruction.
- 1.1.5. **FAS** - Fall Arrest Systems
- 1.1.6. **FPP** - Fall Protection Plan
- 1.1.7. **HIRA** - Hazard identification and Risk assessment
- 1.1.8. **HR** - Human Resources Management
- 1.1.9. **HSE plan** - Health, Safety and Environment Management System
- 1.1.10. **IWH** - Institute for work at Height; the Recognised Professional Body for Fall Protection plan developers in South Africa
- 1.1.11. **MBS** - Minimum Breaking Strain
- 1.1.12. **MEWP** - Mobile Elevated work platform
- 1.1.13. **MFS** - Minimum free space
- 1.1.14. **MSDS** - Material Safety Data Sheet
- 1.1.15. **NQF** - National Qualifications Framework
- 1.1.16. **OEM** - Original Equipment Manufacturer
- 1.1.17. **OHP** - Occupational Health Professional (registered)
- 1.1.18. **OHS Act** - Occupational Health and Safety Act 85 of 1993
- 1.1.19. **PPE** - Personal Protective Equipment (mainly focused on work at height)
- 1.1.20. **PTO** - Planned Task Observation
- 1.1.21. **SANS** - South African National Standards (Previously known as SABS)
- 1.1.22. **SAQA** - South African Qualifications framework
- 1.1.23. **Sec 16(2)** - Assistant to the CEO appointed as per Section 16.2 of the OHS act.
- 1.1.24. **SETA** - Skills Education Training Authorities
- 1.1.25. **SWL** - Safe Working Load
- 1.1.26. **WAH** - Work at height

1.2. Definitions

- 1.2.1. **Anchor Point** - an attachment point at an anchor for anchor lines or persons (SANS50795 or EN 795);
- 1.2.2. **Competent Person** - Competent person for any task a person who has acquired, through training, experience, qualifications and/or experience, or a combination of them, the knowledge and skills to carry out that task, provided that where appropriate qualifications and training are registered in terms of the provisions of the South African Qualifications Authority Act, 1995 (Act No. 58 of 1995), these qualifications and training shall be deemed to be the required qualifications and training.
- 1.2.3. **Design** - In relation to any structure includes drawings, calculations, design details and specifications;
- 1.2.4. **Designer** - any of the following persons-
- (a) a person who prepares a design;
 - (b) a person who checks and approves a design;
 - (c) a person who arranges for any person at work under his control (including an employee of his, where he is the employer) to prepare a design, as well as;
 - (d) an architect or engineer contributing to, or having overall responsibility for the design;
 - (e) building services engineer designing details for fixed plant;
 - (f) surveyor specifying articles or drawing up specifications;
 - (g) contractor carrying out design work as part of a design and building project;
 - (h) any competent person designing false work, including its components, form work and support work;
 - (i) interior designer, shop-fitter and landscape architect.
- 1.2.5. **Energy Absorbing Lanyard** - the connecting element or component of a fall arrest system. A lanyard may be of synthetic fibre rope or webbing. It must have an energy absorbing component incorporated into it, which will keep the impact force below 6 kN shock force in the event of a fall.
- 1.2.6. **Equipment Controller** - A person appointed, in writing, to perform 3 monthly inspection and management, as per the fall protection plan, on all work at height equipment within the organisation.
- 1.2.7. **Exclusion Zones** - An area into which entry is forbidden where tools or equipment can fall off or onto an object or person in the surrounding area.
- 1.2.8. **Fall Arrest** - A form of fall protection which involves the safely stopping a person in the event of a fall. The equipment is designed to stop (arrest) a person's fall in the safest and shortest distance possible.
- 1.2.9. **Fall Arrest Equipment** - equipment used to arrest the person in a fall, including personal equipment such as body harness, lanyards, deceleration devices, lifelines or similar equipment, but excludes single belts.
- 1.2.10. **Fall Arrest and Rescue Technician** - A person who can perform fall arrest manoeuvres, a basic fall arrest rescue and is in possession of valid Fall Arrest and Basic Rescue certificate.
- 1.2.11. **Fall Arrest System** - an assembly of components joined together so that, when connected to a suitable anchor point with sufficient clearance from the ground or other obstacles, it operates as a complete arrangement of equipment able to fulfil its function in use of arresting a fall, e.g. (the safe use of climbing helmet with chin strap (SANS 1397), full body harnesses (SANS 50361, 50358, 50813), energy absorbers (SANS 50355), vertical lifelines (SANS 50353 and SANS 50795, horizontal lifelines(SANS 50353-1 & 2) and SANS 50795), retractable lifelines (SANS 50360) and using a work positioning(SANS 50358) lanyard as body support etc...)
- 1.2.12. **Fall Prevention Equipment** - equipment used to prevent persons from falling from a fall risk position, including personal equipment such as a body harness, lanyards, lifelines or physical equipment such as guardrails, screens, barricades, anchorages or similar equipment;
- 1.2.13. **Fall Protection Plan** - a documented plan of all risks relating to working from a fall risk position, considering the nature of work undertaken, and setting out the procedures and methods to be applied in order to eliminate the risk and must include a rescue plan and procedures;
- 1.2.14. **Fall Risk** - any potential exposure to a risk of falling (Items falling on workers, workers falling from an elevated position or workers dropping items onto other workers). Work cannot continue on site unless this fall risk(s) has been mitigated, reduced and controlled to acceptable levels

- 1.2.15. **Fixed Barricade** - barricading provided to prevent persons from entering areas where there is a Fall risk, falling objects, slipping, open trenches or holes, unprotected edges etc.
- 1.2.16. **Full Body Harness** - body support primarily for fall arrest purposes, i.e. A component of a fall arrest system. Full body harness may comprise straps, fittings, buckles or other elements, suitably
- 1.2.17. **Hazard** - a source of or exposure to danger (source which may cause injury or damage to persons)
- 1.2.18. **Hazard Identification** - the identification and documenting of existing or expected hazards to the health and safety of persons, which are normally associated with the type of construction work being executed or to be executed
- 1.2.19. **Health, Safety And Environmental Plan** - Dedicated health and safety management systems that ensure that appropriate standards of health, safety and wellbeing are maintained by XXX .
- 1.2.20. **Inclement Weather** - Any weather condition that can result in an unsafe work environment. This will be specific to the task and rescue scenario.
- 1.2.21. **Medical Fitness** - being declared medically and fit to work safely from a fall risk position or such similar environment and being in possession of a valid medical certificate of fitness
- 1.2.22. **Medical Certificate Of Fitness** - COF; a certificate valid for one year issued, by an medical health practitioner, whom shall be registered with the Health Professions Council of South Africa: Provided that such certificate will become invalid, temporarily or permanently, subject to a consultation with a health practitioner, if medical or mental conditions occurred or emerged during the period of 12 months (a.k.a. Medical and psychologically certificate of fitness)
- 1.2.23. **Minimum Free Space (MFS)** - Fall arrest system is designed to limit the force on the body during a fall. This is achieved by absorbing the energy generated in the fall, by applying an arresting force to the faller over a distance i.e. The "arrest distance". In order to prevent the possibility of a collision, there must be sufficient free space under the user for the fall to be arrested in i.e. The free space must be greater than the arrest distance. Free space that the path of fall is free from obstacles.
- 1.2.24. **Mobile Elevated Work Platform (MEWP)** - A mobile, raised platform used to provide temporary access for technicians or equipment to inaccessible areas at height.
- 1.2.25. **Night Work** - The period between sunset and sunrise. This is considered to be 15 minutes before sunset and 15 minutes before sunrise.
- 1.2.26. **Permanent Barricade** - barricading which is permanently fixed to prevent falling, or to provide support when ascending or descending stairways using a three points of contact system, such as hand-railing etc.
- 1.2.27. **Portable Ladders** - A temporary structure that can be moved or extended to a desired length. Portable ladders consist of side rails and rungs, there are various types of portable ladders that can be made of various types of materials.
- 1.2.28. **Pre-Task Risk Assessment** - It is a risk assessment that is completed by the site supervisor at the work site before work can start.
- 1.2.29. **Risk Rating** - the probability or likelihood that a hazard can result in injury or damage to property
- 1.2.30. **Risk Assessment** - a process of determining all risks associated with all hazards at a specific construction site, in order to identify the steps needed to be taken to eliminate, mitigate, and or control such hazards.
- 1.2.31. **Rope Access** - a technique using ropes, normally incorporating two separately secured systems, one as a of access and the other as back-up security, used with a harness in combination with other devices, for getting to and from the place of work and for work positioning.
- 1.2.32. **Rope Access Level 1 Technician** - A person who can perform rope access manoeuvres and who can perform a basic rope access rescue and is in possession of a valid Rope Access Level 1 certificate.
- 1.2.33. **Rope Access Level 2 Practitioner** - A person who can perform rope access manoeuvres, supervision and advanced rescues on a simple worksite and is in possession of a valid Rope Access Level 2 certificate.
- 1.2.34. **Rope Access Level 3 Supervisor** - A person who can perform rope access manoeuvres and who can perform a set of advanced rope access rescues on a complex worksite and is in possession of a valid Rope Access Level 3 certificate. A Rope Access Level 3 Supervisor can supervise work on both a simple and complex worksite.

- 1.2.35. **Scaffolding** - A temporary structure used to access a worksite. It is used to support technicians and equipment when building, repairing or cleaning buildings.
- 1.2.36. **Site Supervisor** - A person, appointed in writing, who takes full responsibility of all work at height activities on-site and ensures that all work at height processes are conducted in the correct manner.
- 1.2.37. **Suspended Platform** - a working platform suspended from supports by of one or more separate ropes from each support a.k.a. Suspended access platforms, TSP, BMU etc.;
- 1.2.38. **Toolbox Talks** - communicating the daily hazards and risks on site to the workers and it provides an opportunity to reinforce the safe working procedures and PPE for that specific daily task; it is the most practical step in connecting paperwork and actual work force.
- 1.2.39. **Working Platform** - a platform whose purpose is to support the combined imposed loads of workers, materials and plant. The number of working platforms permitted in simultaneous usage on scaffolding frameworks is limited
- 1.2.40. **Work** - Any task, observation or inspection performed by an Eskom employed person at Medupi power station.
- 1.2.41. **Work at Height** - Any work performed above or below a stable work surface, or where a person puts himself/herself in a position where he/she exposes himself/herself to a fall hazard/risk and they can fall from or into an area that will result in an injury.
- 1.2.42. **Work Positioning** - a technique that enables a person to work supported in tension or suspension by personal protective equipment in such a way that a fall from a height is prevented or restricted
- 1.2.43. **Work Restraint** - a technique whereby a person is prevented by of harness and other devices from reaching zones where the risk of a fall from a height exists

P4. General Site Safety Rules

When on site for training or at work, the following rules must be adhered to at all times:

1. Always wear seat belts when travelling in, or operating vehicles.
2. Always use suitable personal protective equipment, a safety harness and fall protection equipment when working at height (harnesses must be secured at all times when working at height).
3. Never carry out work on electrical equipment, circuits and gear if you are not qualified to do so.
4. Never work under the influence of substances (alcohol or drugs) which are illegal or in excess of legal levels or where this impairs your ability to perform tasks.
5. Never exceed speed limits or travel at speeds which are dangerous for the type of road, vehicle, or conditions.
6. Never use a hand held phone whilst driving and only make calls by pulling over or using hands-free devices, when it is safe to do so.
7. Always wear helmets when travelling on, or operating, a motorbike.
8. No passengers in the backs of bakkies and trucks.
9. Never undertake any street or underground work activities unless competent to do so.



These rules have to be implemented at all times regardless of work being done.

1. Introduction to “Just Culture”:

Notes:

Just culture supports the idea of learning from unsafe acts wherever they may be observed in order to improve safety and safety awareness through the improved recognition of safety situations.

* The “Just Culture” duties and responsibilities of all workers in a fall risk area:

Competent persons and employees have accepted the fact that no fall risk environment can be made completely risk free.

Therefore, in order to implement an FPP, all employees and the company need to ensure that we have **completed the risk controls correctly**.

We all have to accept the responsibilities that the following duties require of us in order to reduce fall risks to a reasonable and practicable safe level:

1. I have a duty to avoid causing unjustified risk or harm during the task.
2. I have a duty to produce an outcome.
3. I have a duty to follow a procedural rule (SOP etc)*.

*See pre-task risk assessment for the declaration and procedure that confirms the above policy.

Inattention Blindness:

Psychology research has uncovered an interesting phenomenon that relates to the “I-didn’t-see-you” excuse. It seems that humans are subjected to so much incoming information that one part of the brain serves as sort of a subconscious mental “spam filter.” The filter passes on to the conscious part of the brain only information that corresponds to what the person is concerned about, or what the psychologists call attending to. If a driver is thinking about (attending to) buying a new car, the spam filter will pass along anything related to cars, including automobile dealerships and billboards with car ads. The flip side is that if the driver is not concerned about something (say, motorcycles), the spam filter may delete the image. So it really can be true that a driver looks right at you and sees you, but your image gets filtered out and never gets to the conscious part of the driver’s brain. That would explain how a driver might not comprehend a motorcyclist wearing a high-visibility jacket and the bike flashing its headlight—or, for that matter, a freight train with a flashing headlight and the air horns blaring.

Thus, it is important that the worker conditions him/herself to be attentive to any and all factors that may influence their safety or the safety of those around them. To enter a worksite with the mentality that safety is the highest priority and must therefore be at the forefront of the conscious mind.

2. Requirements for safely Performing Work in Elevated Positions

Notes:

Before a Fall Protection Plan (FPP) is written it is important to note that an FPP is not a stand-alone document, but rather a document that forms part of, and compliments the current Health, Safety and Environmental (HSE) Plan. Thus, it must be remembered, when writing an FPP, to not contradict or overrule any HSE policies or procedures. Furthermore, this course will address fall risks and the requirements to safely mitigate those risks which are not typically included in an HSE Plan.

The Occupational Health and Safety Act 85 (OHS Act of 1995), addresses the requirements of how to perform work safely. This includes work at height, not only Fall Arrest and Rope Access, but ANY work where a person may fall. This is referred to as a Fall Risk Position. In order to mitigate these fall risks, the OHS Act contains Construction Regulations (CR's), in particular CR 10, which requires a Fall Protection Plan. The purpose of this plan is to control any and all fall risks as far as reasonably practical. It is the purpose of this course to instruct the learner on how to implement this plan.

2.1 CR 10(1) A contractor must-

10(1)(a) designate a competent person to be responsible for the preparation of a fall protection plan;

A competent Fall Protection Plan Developer must ensure that the entire Fall Protection Plan must be completed before any work may start.

Thus, it must be understood what the requirements of a Fall Protection Plan is and by extension the general requirements for performing work at height. These requirements are given in Construction Regulations 10(2).

2.2 CR 10(2) A Fall Protection Plan contemplated in sub-regulation (1), must include-

10(2)(a) a risk assessment of all work carried out from a fall risk position and the procedures and methods used to address all the risks identified per location;

10(2)(b) the processes for the evaluation of the employees' medical fitness necessary to work at a fall risk position and the records thereof;

10(2)(c) a programme for the training of employees working from a fall risk position and the records thereof;

10(2)(d) the procedure addressing the inspection, testing and maintenance of all fall protection equipment; and

10(2)(e) a rescue plan detailing the necessary procedure, personnel and suitable equipment required to affect a rescue of a person in the event of a fall incident to ensure that the rescue procedure is implemented immediately following the incident.

2.3 CR 10(3) - A contractor must ensure that a construction manager appointed under regulation 8(1) is in possession of the most recently updated version of the Fall Protection Plan.

3. Planning for Work at Height

Notes:

3.1 CR 10(2)(a) - Risk Assessment

Before work starts, the Client must have provided the Contractor with a "baseline risk assessment" that should cover all risks prevalent on site as per CR 5. This baseline risk assessment must be taken into account by the Fall Protection Plan Developer when performing the work at height risk assessment, which will be the risk assessment included in the Fall Protection Plan as per CR 10(2)(a). The risk assessment must be conducted as per CR 9.

CR 9 - CR9(1) A contractor must, before the commencement of any construction work and during such construction work, have risk assessments performed by a competent person appointed in writing, which risk assessments form part of the health and safety plan to be applied on the site, and must include-

- 9(1)(a) the identification of the risks and hazards to which persons may be exposed to;
- 9(1) (b) an analysis and evaluation of the risks and hazards identified based on a documented method;
- 9(1) (c) a documented plan and applicable safe work procedures to mitigate, reduce or control the risks and hazards that have been identified;
- 9(1) (d) a monitoring plan; and
- 9(1) (e) a review plan.

It is the duty of the Fall Protection Plan Developer to ensure the following is identified:

- **The prevalent risks on site:**
 - Falling from height.
 - Workers dropping tools or equipment from height.
 - Tools or equipment falling onto people below.
 - Dangerous areas where people are not allowed etc.
- **The risk controls established for the above mentioned risks:**
 - What are the needed steps to be taken in order to reduce the risks to an acceptable level. The implementer has to identify the risks and ensure that the relevant baseline risk assessment controls are implemented to ensure a safe work environment. This is indicated on the pre-task risk assessment
- **The risk owners:**
 - The person responsible for ensuring the risk controls/mitigations are implemented and continuously adhered to. The risk owners may vary from the employer to the employee, depending on the risk and the controls

All risk assessment training must be recorded and the records must be kept.

All of the above is accomplished with induction training and Daily Safe Task Instructions or pre-task risk assessments. It is recommended that DSTI be monitored at least weekly by a competent person(CR8(5), CR8(8), CR17(2)).

3.2 Notes on a Work at Height Risk Assessment

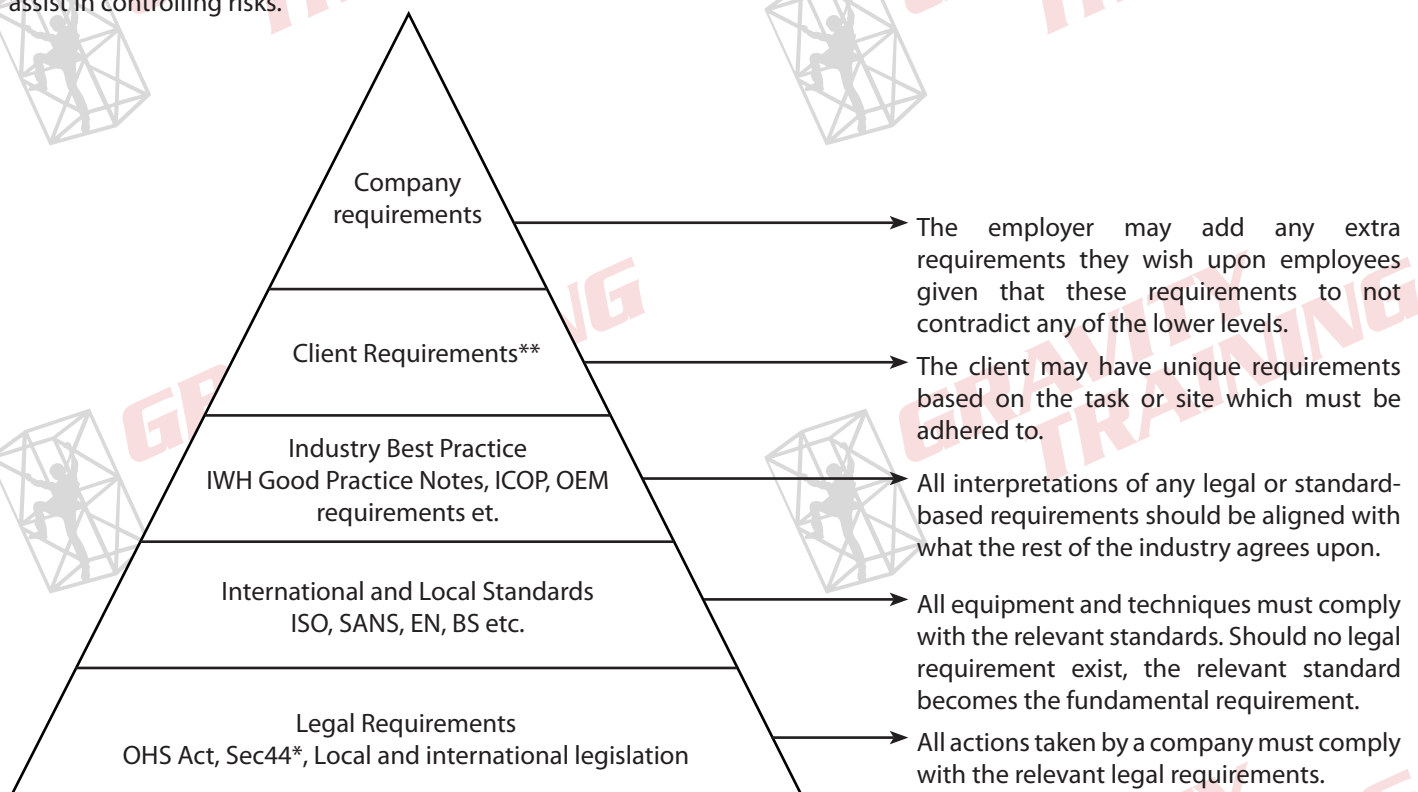
The work at height risk assessment forms the basis of the Fall Protection Plan, and as such it is of utmost importance that it is written in such a way that allows a company to establish and maintain a safe work environment with as little effort as possible.

Thus, when writing a work at height risk assessment, place extra emphasis on the following:

- The work at height risk assessment must be written by a competent person.
- Ensure that as many hazards and risks are identified as possible in order to be able to prevent any surprises when a site is established.
- Risk mitigations must be practical and reasonable as far as possible.
- Risk mitigations must not introduce any new risks onto site. If a risk mitigation does introduce new risks (for example: using Fall Arrest), they must be mitigated similar to all other risks.
- Ensure that the process of assigning risk ratings is documented (risk matrix) and understood by all persons involved in the risk assessment process.
- Beware of writing impossible-to-implement risk mitigations.
- Beware of writing mitigations in such a way that the company becomes liable for any incidents; for example, stating that workers will stop work to take lunch at 12:00, if a worker were to fall at 12:05, the company becomes vulnerable. It would be better to write the same mitigation, but replacing the "12:00" with "as needed".
- An Issue-Based Risk Assessment must be conducted per site and task to ensure the relevant risks on site are identified and mitigated as per site and task. This is to be completed by the Fall Protection Plan Developer before work starts.

3.3 Risk Management Hierarchy

Below is a diagram showing the hierarchy that must be followed when dealing with risk controls or procedures designed to assist in controlling risks.

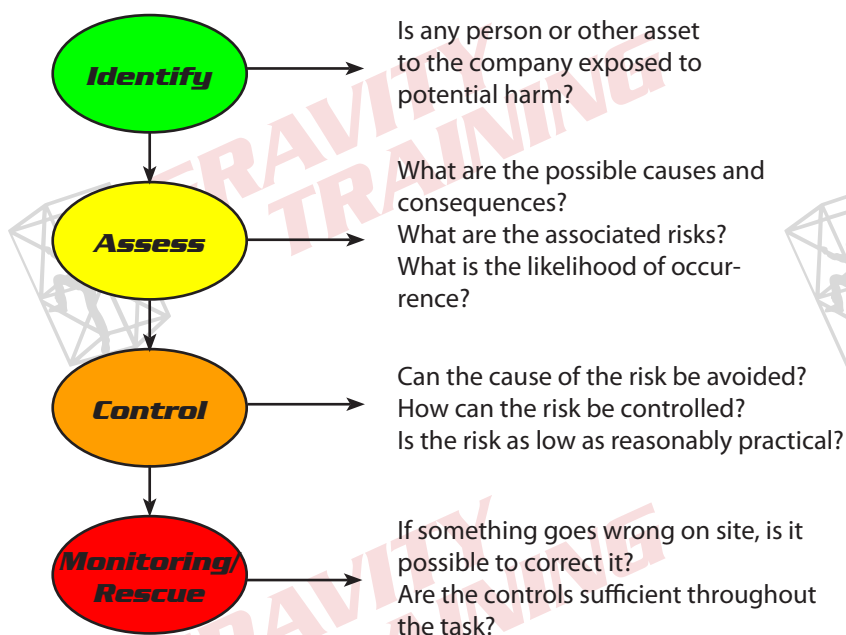


*Sec44 - Section 44 of the OHS Act states that the minister may "promote" any standard to a legal requirement under the allowances made in Section 44 of the OHS Act

**Client Requirements - When the Sec37(2) agreement is signed between the client and the contractor, all client requirements become legal requirements.

3.4 Risk Management Flowchart

Below is an illustrated example of the process of risk management.



When a risk is identified, it needs to be analysed and given a Risk Rating - it is given a rating twice - once before any control methods have been implemented, and once afterwards. In order for us to consistently analyse our identified risk, and ensure that we can reproduce any rating, we need a "documented method" for this process as stated in CR 9(1)(c).

No work is allowed to commence if there is a **"High"** risk present on site.

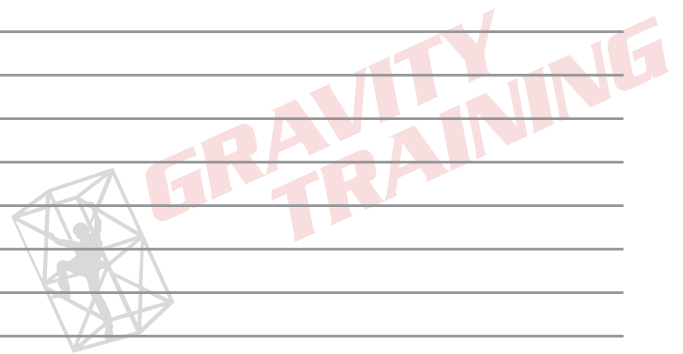
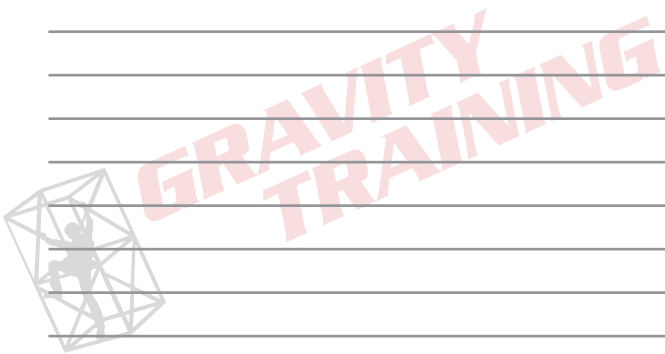
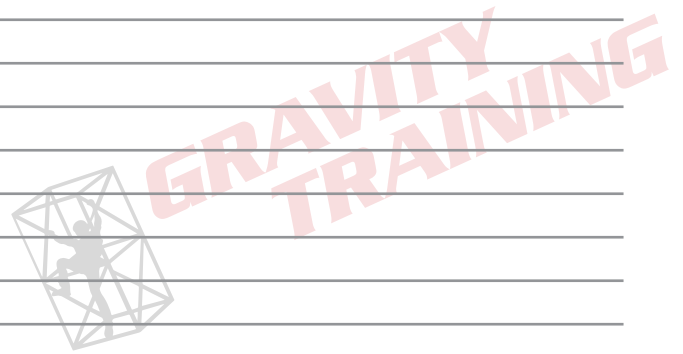
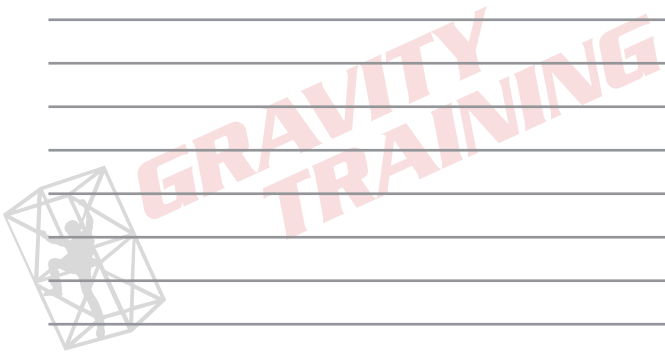
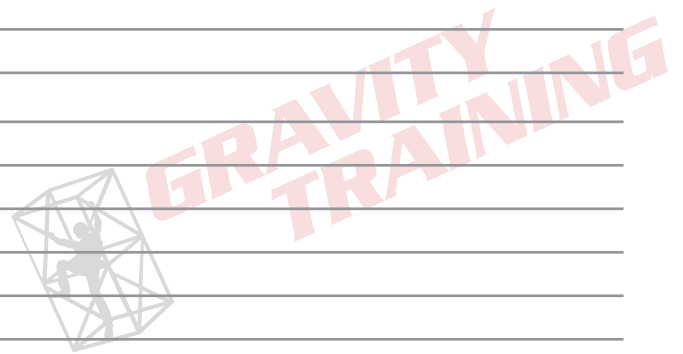
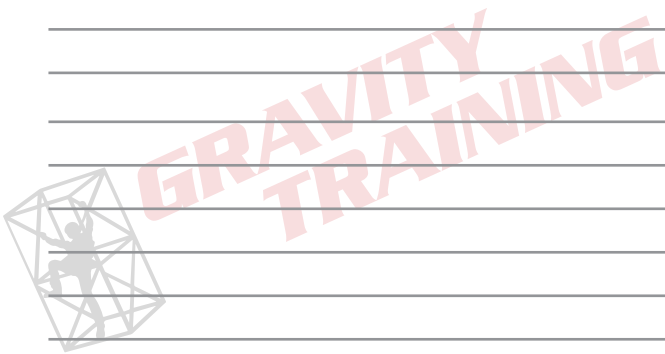
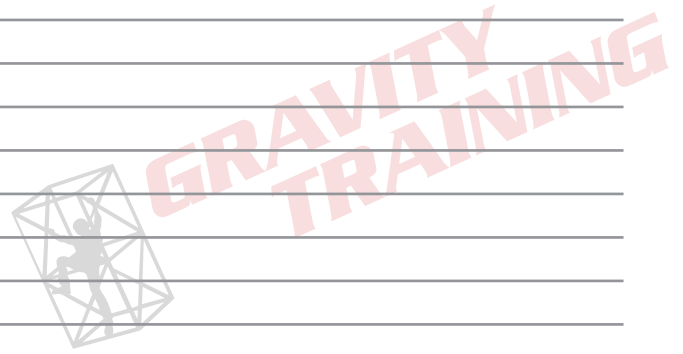
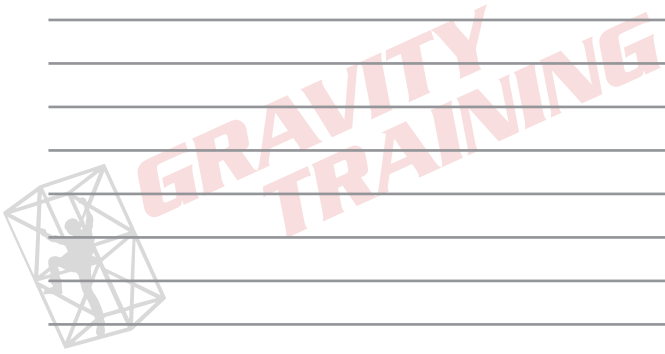
Thus, we have to mitigate all identified risks in order for them to receive a **"Low"** risk rating. **"Medium"** are acceptable as long as a SME has been consulted.

3.5 Risk Matrix

Risk Matrix			Hazard Severity				
			1	2	3	4	5
			Minor Injury, no time off work	Injury resulting in up to three days of work	Injury resulting in more than three days off work	Major disabling injury (e.g. loss of a limb or an eye)	Fatality
Likelihood	1	Highly unlikely to occur	1. Low	2. Low	3. Low	4. Low	5. Low
	2	Possibility remote but has occurred	2. Low	4. Low	6. Low	8. Medium	10. Medium
	3	Very infrequently	3. Low	6. Low	9. Medium	12. Medium	15. High
	4	Occasionally	4. Low	8. Medium	12. Medium	16. High	20. High
	5	Frequently and regularly	5. Low	10. Medium	15. High	20. High	25. High

Risk value result based on Risk Matrix	Recommended Action
Low(1-6)	May be acceptable; however, review the task to see if the risk can be reduced further.
Medium (8-12)	Where possible, the task should be redefined to take account of the hazards involved or the risk should be reduced further, prior to task commencement. Appropriate management authorisation after consultation with specialist personnel and an assessment team may be required.
High(15-25)	Unacceptable. The task should be redefined or further control measures put in place to reduce the risk. The controls should be re-assessed for adequacy, prior to start of task.

Notes:



3.6 Example of a Work at Height Risk Assessment

Notes:

Hazard	Risk	Pre-Mitigation Risk Rating			Mitigations	Post-Mitigation Risk Rating		
		Likelihood	Severity	Risk Rating		Likelihood	Severity	Risk Rating
1. Person at height (exposed to fall risk)	1.1 Person unaware of risks on site				1.1.1 A work at height risk assessment must be conducted by a competent person before work starts.			
					1.1.2 All workers must be inducted on the work at height risk assessment by a competent person before work starts.			
					1.1.3 A daily risk assessment must be conducted and all persons on site must be inducted by a competent person before work starts/continues.			
					1.1.4 All inductees must sign an induction register to prove that they have been made aware of the risks on site as well as the relevant mitigations, these registers must be kept.			
		4	4	H		1	4	L
	1.2 Incompetent supervision	3	5	H				
	1.3 Incompetent workers	5	5	H				
2. Object at height (exposed to drop risk)								
3. Other								

3.6.1 Hazards

Hazard
1. Person at height
2. Object at height
3. Other

A hazard is defined as an event that may lead to harm, damage to property, injury and/or death. Hazards cannot be mitigated, however, it may be made safe with sufficient mitigation of the risks that a given hazard may pose.

Person at height - A person at height is the main hazard for a work at height environment, thus the risks that may lead to a fall of a person, must be mitigated sufficiently.

Object at height - Objects at height pose a risk of injury and/or death not only to the workers on site, but to any members of the public that may be in the area. As such, the risks that may lead to a person dropping objects must be strictly controlled in order to create a safe work site.

Other - Any other hazards that are identified that will affect the safety of workers or members of the public must be controlled and documented. For example: RF radiation, electrical hazards, traffic etc.

3.6.2 Risks

Risk
1.1 Person unaware of risks on site
1.2 Incompetent supervision
1.3 Incompetent workers

A risk is the cause or trigger that may lead to damage or a person being harmed or injured within the hazardous environment.

Person unaware of risks on site - A person who is unaware of the risks on site will undoubtedly also be unaware of what actions to take to avoid the risk and will thus place themselves and/or others in harms way.

Incompetent supervision - A person who is unqualified to supervise a given team and/or task, will be unable to identify "at-risk"-behaviour. This will thus place the team members, and possibly members of the public, in danger.

Incompetent workers - Workers who are not competent for a given task will not be able to perform the task safely due to them not adhering to the correct principles/procedures, which can lead to them injuring themselves or others.

3.6.3 Pre-mitigation Risk Rating

Pre-Mitigation Risk Rating		
Likelihood	Severity	Risk Rating
4	4	H

The pre-mitigation risk rating is based upon the risk matrix given on page 5 and refers to the state of the risk BEFORE ANY CONTROL MEASURES HAVE BEEN IMPLEMENTED. The rating is calculated as follows: firstly, the likelihood is considered on a scale from A (very unlikely) to E (almost certain). Then the severity is considered on a scale from 1 (insignificant) to 5 (catastrophic). For example:

4 - Occasionally - There is not only a chance of an occurrence but a chance of frequent (more than once) occurrences.

4 - Major Disabling Injury - A high severity that indicates that, should the risk remain uncontrolled, serious injury (loss of limb or eye etc) will occur.

Risk Rating - If we combine these two factors, the likelihood that a person will be injured or harmed and the fact that that injury or harm will be serious, using the risk matrix, we arrive at the final pre-mitigation risk rating of HIGH, which indicates that no work may commence or continue without additional controls being implemented. As shown below.

Risk Matrix		Hazard Severity				
		1	2	3	4	5
Likelihood	1	1. Low	2. Low	3. Low	4. Low	5. Low
	2	2. Low	4. Low	6. Low	8. Medium	10. Medium
	3	3. Low	6. Low	9. Medium	12. Medium	15. High
	4	4. Low	8. Medium	12. Medium	16. High	20. High
	5	5. Low	10. Medium	15. High	20. High	25. High

Notes:

3.6.4 Mitigations

After all the risks have been identified and evaluated, the risks can be prioritised as per their risk ratings (from high to low) so that the most pressing risks are mitigated first.

Mitigations
1.1.1 A work at height risk assessment must be conducted by a competent person before work starts.
1.1.2 All workers must be inducted on the work at height risk assessment by a competent person before work starts.
1.1.3 A daily risk assessment must be conducted and all persons on site must be inducted by a competent person before work starts/continues.
1.1.4 All inductees must sign an induction register to prove that they have been made aware of the risks on site as well as the relevant mitigations. These registers must be kept.

Mitigations are the control measures put in place that must be obeyed by everyone on site in order to reduce the risks and create and maintain a safe work environment. All mitigations must contain the following:

A risk owner (who) - The person who will be responsible for ensuring that the risk is controlled.

List of most common legal appointments:

- CR 8(1) - Construction Manager
- CR 8(5) - Safety Officer
- CR 8(7) - Site Supervisor
- CR 9(1) - Risk Assessor
- CR 10(1)(a) - Fall Protection Plan Developer
- CR 16(1) - Scaffolding Supervisor
- CR18(1)(a) - Rope Access Supervisor
- Sec 16(1) - CEO
- Sec 16(2) - Assistant to CEO

An action (what) - What action must be taken by the risk owner to ensure that the risk is controlled.

A time (when) - When must action be taken by the risk owner to ensure that the risk is controlled.

A risk assessment must be conducted before work may start, this risk assessment must not be site-specific and must cover all risks that may possibly be faced on a work at height site.

The risk assessment will only be effective if the relevant persons are aware of the controls that need to be implemented. However, it is currently Industry Best Practise to induct all workers every time a risk assessment changes. [All workers must be re-inducted annually.

Since the risks that are prevalent on site may change frequently, the site must be evaluated daily before work starts to ensure that all risks have been identified and controlled as per the work at height risk assessment. Should a visitor or any other person, who was not inducted on the original daily risk assessment, arrive on site, this person must be inducted on the risk assessment as well. This does not differ based on the person's position within the company.

Proof must be kept that all relevant persons have been inducted on the risk assessments. Thus, induction registers must be signed and maintained.

Notes:

3.6.5 Post-mitigation Risk Rating

Post-Mitigation Risk Rating		
Likelihood	Severity	Risk Rating
1	4	L

The pre-mitigation risk rating is based upon the risk matrix given on page 5 and refers to the state of the risk BEFORE ANY CONTROL MEASURES HAVE BEEN IMPLEMENTED. The rating is calculated as follows: firstly the likelihood is considered on a scale from A (very unlikely) to E (almost certain). Then the severity is considered on a scale from 1 (insignificant) to 5 (catastrophic). For example:

→ **A - Highly Unlikely to occur** - May occur, but only in rare and exceptional circumstances.

→ **4 - Major Disabling Injury** - A high severity that indicates that, should the risk remain uncontrolled, serious injury (loss of limb or eye etc) will occur.

→ **Risk Rating** - If we combine these two factors, the likelihood that a person will be injured or harmed and the fact that that injury or harm will be serious, we arrive at the final pre-mitigation risk rating of HIGH, which indicates that no work may commence or continue without additional controls being implemented.

Risk Matrix		Hazard Severity				
		1	2	3	4	5
Likelihood	1	1. Low	2. Low	3. Low	4. Low	5. Low
	2	2. Low	4. Low	6. Low	8. Medium	10. Medium
	3	3. Low	6. Low	9. Medium	12. Medium	15. High
	4	4. Low	8. Medium	12. Medium	16. High	20. High
	5	5. Low	10. Medium	15. High	20. High	25. High

Notes:

Example of a Pre-task Risk Assessment (Complete this for Site 2) :

Site Area		Date	
Site Task Area			
Permit number(s)			
Work at Height Risk Assessment No			
Supervisor Name		Contact No	
SCOPE OF WORKS			
Describe task to be performed:			
List the relevant steps of the task			
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
First Aider/Rescuer:	1:	2:	3:
EMS No:		Police No:	
Fire/Rescue Team No:		Company Contact Details	
How will EMS access site:			
SITE SPECIFIC RESCUE PLAN			
Rescue method	☐ First Aid Kit	☐ Rope Grabs	☐ Descending Device
	☐ Ratchet Rescue Kit	☐ Rescue Slings	☐ RA lift and lower kit
	☐ Stretcher and basket	☐ Plug-in for tensioned lines	☐ RA lowering kit
	☐ Cableway rescue kit		☐ Rig for rescue
Note CR8(7) and/or competent rescuer on site to discuss the rescue procedure on site as per the task requirements			

NO.	TASK/HAZARD	MITIGATION	TICK
1.	☐ Task specific risks/hazards	Refer to HSE plan. CR8(7) to ensure that all risks/hazards on site are managed reasonably and practicably including HSE concerns.	
2.	☐ Medically fit to perform task	The Site Supervisor to ensure that all workers are in possession of a valid C.o.F before work starts	
3.	☐ Rope Rigging	CR8(7) to ensure that the correct SOP/Method Statements are followed as per training. OEM and industry practises to be followed. Constant competent supervision is required. A lifting plan must be completed and implemented	
4.	Work Platform ☐ Rooftop ☐ Fragile Surfaces ☐ Open Edges ☐ Open Holes ☐ Hand railings/parapet wall or crow's nest ☐ Sloping edges ☐ Container ☐ Adequate safe anchors/fall arrest systems Other:	CR8(7) must ensure that the work platform is safe for use and fit for purpose. Appropriate fall arrest/work restraint systems to be used where required as per training and OEM requirements.	
5.	Access Method(s) ☐ Lattice towers ☐ Portable ladders ☐ Fixed or caged ladders ☐ Scaffolding ☐ MEWPs ☐ Man cage/cradle ☐ Monopoles ☐ Rope Access Other:	Site Supervisor to ensure that all workers on site have been inducted on the baseline risk assessment and are aware of the correct mitigation to ensure a safe work area is maintained All access methods to be used as per the latest version of the FPP and relevant SOP's. CR8(1) to ensure competent supervision is appointed as per relevant Construction Regulations and chosen access methods throughout the task.	
6.	PPE Inspection ☐ PPE Usage (competence and safe use) ☐ Pre-use inspection ☐ Long-term inspection	A competent appointed person must inspect all PPE as per OEM and industry best practises. The user must ensure to inspect their PPE before use with the use of the buddy system	
7.	☐ Equipment Inspection	The user must ensure to inspect his equipment before use with the use of the buddy system	
8.	☐ Environmental	CR8(7) to ensure that no work is conducted in adverse weather conditions or any other unsafe environmental conditions and employees to report any concerns to the CR8(7).	
9.	☐ Dropping Tools/Items	CR8(7) to ensure that the correct barricading is in place and all tools/items are to be tied off. Helmets must be worn by all persons in a dropzone	
10.	☐ Unprotected Edges	CR8(7) to communicate unprotected edges to all workers before work starts and to supervise unprotected edges at all times. CR8(7) to ensure proper barricading is in place where possible. Workers can make use of work restraint or fall arrest.	

11.	Sharp/unsafe edges	CR8(7) to ensure no PPE or other equipment are used near/over sharp edges hazardous area. If PPE or other equipment are close to sharp edges or any other hazardous area edge protection must be used at all times	
12.	Access/exclusion zones	☐ Working edge hazard zone ☐ Top exclusion zone ☐ Exclusion zone at intermediate level ☐ Bottom exclusion zone ☐ Rigging anchor exclusion zone ☐ Patrol officer required Name and Number:	
13.	Uninformed workers	Site Supervisor to ensure that all workers on site have been inducted on the baseline risk assessment and are aware of the correct mitigations to ensure a safe work area is maintained	
14.	Suspension trauma	Site Supervisor and rescuer to ensure that a site specific rescue plan is completed and any rescue to be performed as quickly and efficiently as possible	
15.	Other:		
16.	Other:		
17.	Other:		
18.	Other:		
19.	Other:		
20.	Other:		

DAILY SAFE TASK INSTRUCTION ON SITE

Confirmation to Risk Assessment during each Toolbox Talk:

1. I hereby confirm that I have completed the required induction training program.
2. I commit to ensure all work done on a work site will be done in a safe environment and will follow the required safe use procedures.
3. I also commit to immediately notify my employer and or site supervisor immediately if any task cannot be performed safely or if you become aware of any unsafe condition or action. You also commit to ensure that the risks will be safely and correctly mitigated before work continues on site.
4. I commit to protect myself and those around me.
5. I confirm that I will inspect all PPE and any other equipment before use with the help of the Buddy System.

Name and Surname	Designation	Signature	Date
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

3.6 CR 10(2)(b) - A process for determining medical fitness of workers

Notes:

Part of the Risk Assessment has to address the risk of falling from height where one of the causes of this could be due to medical unfitness. Thus there must be a process for determining whether a person is medically fit to perform the specified work, as well as having proof of said medical fitness based on the persons' job specifications (driver, welder, work at height technician etc.). This usually comes in the form of a Certificate of Fitness (COF) in the format of Annexure 3 of the Construction Regulations. The Certificate of Fitness must contain the following details:

- The employee's full name and surname.
- The employee's ID number.
- The date of the examination.
- The date of expiry.
- The result of the examination; fit or unfit as per job specification.
- Must be signed off by an Occupational Health Practitioner.

In the same way that the risk assessment may be subject to near constant change, so can the medical fitness of workers be subject to rapid and unpredictable change, based on the site and the scope of work. Thus, it is not sufficient to rely solely on the COF as proof of medical fitness, but to add the extra precaution of daily discussions with team supervisor to ensure that no short-term medical conditions/sickness or general unwellness are present that may affect the safety of the team. a buddy system may also be employed to assist in managing short-term medical concerns.

It might be required that workers are subject to more frequent medical evaluations. This may be due to confined space work, toxic gasses or hazardous chemicals.

The Fall Protection Plan must include the correct procedure relevant to the site and scope of work. It is the duty of the Fall Protection Plan Implementer to ensure that this procedure is followed by everyone on site and that the COF's are available on request, as well as any other required proof of this implementation. It is also recommended that a COF register is kept and managed by an appointed responsible person (HSE representative, 8(5), 8(7) etc.) with the dates of the COF expiry date as well as the date of the next examination.

Note: The results of the medical evaluation must be kept for future reference, however these results are confidential and may either be kept the employees personnel file or at the OHP's office.

Below is an example of Annexure 3.

Name of Employee	ID Number	Co Number
-------------------------	------------------	------------------

[illegible]

"The employer to complete the information in the space marked with an * before sending the Employee for a medical examination

Declaration by the Medical Examiner

I certify that I have, by examination and testing, using the above criteria specified by the employer, satisfied myself that the above named employee is fit to perform the duties as described by the employer in the notice above.

Compositional Medicine: Practitioner/Compositional Health Nursing Practitioner. (Please Print Name)

Signature _____
Practice Number _____
Date _____

Address _____

3.8 CR 10(2)(c) - A Training programme

Notes:

To ensure that all workers on site are aware of the correct method to use in order to perform a given task safely, workers must undergo training.

The required training is not limited to Fall Arrest or Rope Access training, but includes any and all training required for a worker to perform his/her task effectively, competently and, above all, safely.

The Fall Protection Plan will include a "training programme" which will stipulate the training required as per job specification, as well as when and how often the training must be received.

Typical Work at Height training includes, but is not limited to:

- Fall Arrest - US 229998
- Basic Fall Arrest Rescue - US 229995
- Advanced Fall Arrest Rescue - US 229999
- Rope Access Technician - US 230000
- Rope Access Practitioner - US 229996
- Rope Access Supervisor - US 229997 and US 230001
- Rope Rigging - US 14706
- Mobile elevated work platforms
- Temporary suspended platforms
- Scaffolding

Apart from access method specific training, all workers must also receive training on the Fall Protection Plan, or what is referred to as FPP Induction. This includes:

- Risk Assessment.
- Emergency Procedures.
- Safe Operating Procedures.

The Fall Protection Plan will include the frequency at which these trainings must be conducted; some might be every 3 years, some might be required more than once a day. It is the duty of the Implementer to ensure that there is proof that the required training has been conducted at the required intervals as set forth by the Fall Protection Plan.

Proof of all training must be available on request, this may be given in the form of certificates, induction registers, or attendance registers (where proof of attendance and not competency is required)

On the next page is an example of a "training matrix" which specifies the typical yearly training requirements as per typical designations.

Fall Risk Training Matrix														Annexure 1				
No	Designation:	Training requirements per designation from a IWH and SETA approved training Provider (Trainer and Assessor to be a “Competent Person(s)”)																
		FPP Induction	FPP Implementation	98	95	99	94	06	CEIM	00	96	01/97	FA	Other				
1.	Assistant to CEO	Yes	-	-	-	-	-	-	-	-	-	-	-	-				
2.	FPP Developer	-	-	Yes	Yes	-	Yes	-	-	-	-	-	-	-				
3.	FPP Implementer (at least 2 x per region)	-	Yes	Yes	Yes	-	-	-	-	-	-	-	-	-				
4.	Construction Manager	Yes	-	Yes	-	-	-	-	-	-	-	-	-	-				
5.	Construction Supervisor	-	Yes	Yes	Yes	-	-	-	-	-	-	Yes	-	-				
6.	Equipment Controller (at least 2 x per region)	Yes	-	Yes	Yes	-	-	-	Yes	-	-	-	-	OEM Specific training				
7.	Basic Fall Arrest Technician	Yes	-	Yes	-	-	-	-	-	-	-	-	-	-				
8.	Fall Arrest & Basic Rescue Technician	Yes	-	Yes	Yes	-	-	-	-	-	-	Yes	-	-				
9.	Advanced Fall Arrest Rescue Technician	Yes	-	Yes	Yes	Yes	-	-	-	Yes	-	Yes	-	-				
10.	Rope Rigger	Yes	-	Yes	Yes	-	-	Yes	-	-	-	-	-	-				
11.	Man-cage Operator	Yes	-	Yes	Yes	-	-	-	-	-	-	Yes	-	Man-cage specific operating license as per DMR and OEM.				
12.	MEWP User	Yes	-	Yes	-	-	-	-	-	-	-	-	-	-				
13.	MEWP Operator	Yes	-	Yes	Yes	-	-	-	-	-	-	Yes	-	US 243273 – Monitor and control the safety and operations of MEWPs				
14.	Suspended Platform Supervisor	Yes	-	Yes	Yes	-	-	-	-	-	-	Yes	-	US 243271 – Supervise, inspect and control the use of temporary suspended platforms				
15.	Scaffold Supervisor	Yes	-	Yes	Yes	-	-	-	-	-	-	Yes	-	US 263234 – Supervise the erection and dismantling of access scaffolding.				
16.	Rope Access Technician	Yes	-	Yes	-	-	-	-	-	Yes	-	-	-	-				
17.	Rope Access Practitioner	Yes	-	Yes	-	-	-	-	-	Yes	Yes	Yes	-	-				
18.	Rope Access Supervisor	Yes	-	Yes	-	-	-	-	-	Yes	Yes	Yes	Yes	-				

*FPP Induction must be conducted each and every time the Fall Protection Plan changes.

The competencies and qualifications that are obtained as per the Training Matrix can be recorded in a Training Register (FPPCD03_01), this will simplify the management of the training.

Legend:

- 98 – US 229998 - Basic Fall Arrest
- 95 – US 229995 - Basic Fall Arrest Rescue
- 99 – US 229999 - Advanced Fall Arrest Rescue
- 94 – US 229994 - Fall Protection Plan Developer
- 06 – US 14706 - Rope Rigging
- CEIM – Climbing Equipment Inspection and Management
- 00 – US 230000 - Level 1 Rope Access Technician
- 96 – US 229996 - Level 2 Rope Access Practitioner
- 01/97 – US 230001 & 229997 - Level 3 Rope Access Supervisor
- FA – First Aid

Technician Name	Technician Name	Technician Name	Technician Name
	Cert Expiry Date	Cert Expiry Date	Cert Expiry Date
Administrative			
ID			
Passport			
OHP Medical (Annexure 3)			
Defensive Driving Course			
Work at Height Training			
US 229998 (Basic Fall Arrest)			
US 229995 (Fall Arrest and Basic Rescue)			
US 229999 (Advanced Fall Arrest Rescue)			
230000 (Level 1 Rope Access)			
229996 (Level 2 Rope Access)			
US 229997 & 230001 (Level 3 Rope Access)			
US 14706 (Rope Rigging)			
US 229994 (Fall Protection Plan Developer)			
Climbing Equipment Inspection and Management			
Basic Ladder			
Working in Confined Spaces			
Mechanical Lifting			
IRATA Rope Access Level 1			
IRATA Rope Access Level 2			
IRATA Rope Access Level 3			
Artisan Rigging			
Advanced Aerial Rigging			
Health and Safety			
SHE Representative			
HIRA			
Incident Investigation			
Basic Fire Fighting			
First Aid			
Legend			
Still Valid			
Will expire in the next 3 months			
Already Expired			
Expired			
Training Booked			
Pending - waiting for certificate			

Above is an example of a training register which is given on the Gravity Training 229994 CD.

In the past, there has been some confusion as to when Unit Standard training is required, compared to when in-house training is acceptable. Again we refer to the OHS Act:

The Construction Regulations defines a "competent person" as someone who :

"has in respect of the work or task to be performed the required knowledge, training and experience and, where applicable, qualifications, specific to that work or task: Provided that where appropriate qualifications and training are registered in terms of the provisions of the National Qualifications Framework Act, 2000 (Act No. 67 of 2008), those qualifications and that training must be regarded as the required qualifications and training."

NQF - According to this definition, if a course is registered as a Unit Standard with SAQA, any person found competent against that standard will be deemed as competent for the given work or task.

LNQ - Should there not exist a Unit Standard for a given qualification or task, a course may be registered with the Institute for Work at Height (IWH), as a Learning Non-Qualifications (LNQ). This provides the course with the credibility that accompanies IWH accredited courses and ensures that a high level of quality training is maintained.

Non-NQF - Should no Unit Standard exist, and no LNQ has been registered, competence in a non-NQF course will be considered as proof of competence for a given task or work, provided that this course has been presented by a reputable training provider.

10(2) (d) the procedure addressing the inspection, testing and maintenance of all fall protection equipment.

Referring back to the risk assessment, another cause of people, tools or equipment falling might be equipment failure. To prevent this, there must be a procedure in place for the inspection, testing and maintenance of equipment. This must be done by a person who is competent not only in Climbing Equipment Inspection and Management, but also in Fall Arrest, to be able to ascertain the suitability and safety of the equipment for a given task.

All equipment used on site must be inspected at intervals specified by the manufacturer or the relevant standards. ALWAYS FOLLOW OEM REQUIREMENTS FOR EQUIPMENT.

According to ISO 22846 work at height equipment must be inspected every 6 months, before, after and during use, as well as daily and interim inspections are needed. However it is considered industry best practice to inspect equipment every 3 months. The relevant inspection criteria, storage requirements and replacement procedures of equipment is given here for the benefit of the Fall Protection Plan Developer in order to provide a basis for the developer to be able to make a judgement on the sufficiency of the equipment management.

All requirements, policies, procedures, and responsibilities for equipment management must be documented and recorded in the Fall Protection Plan and it is the duty of the Fall Protection Plan Implementer to ensure they are implemented.

Inspection Criteria:

	
Software	Hardware
• Condition of stitching • Discolouration • Wear and tear • Cuts • Burns • Date of manufacture • (for rope) Condition of core	• Rust • Deformation • Working action • Movement • Spring action • Wear and tear

It is important to note that, although the Construction Regulations call for an equipment testing procedure, no PPE may be subject to load test, drop test, or any form of destructive testing and then subsequently be used again.

Should companies wish to test equipment, refer all requirements to the OEM and record all relevant procedures in the Fall Protection Plan.

3.12 Equipment Storage

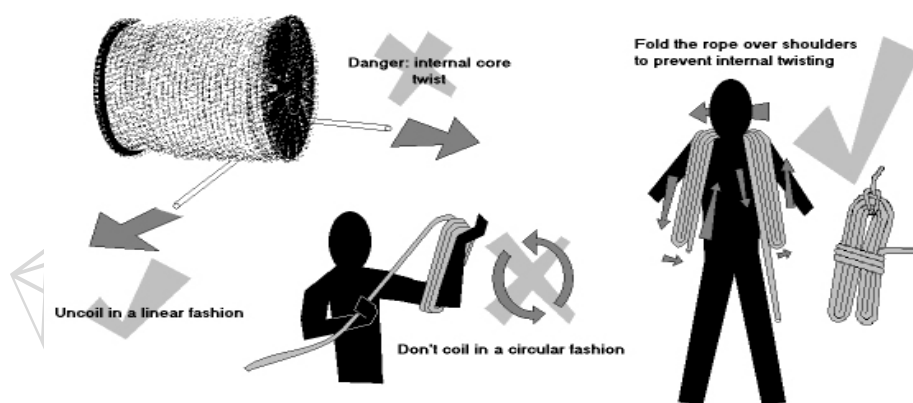
Notes:

Equipment must be stored correctly and according to manufacturer's specifications to prevent damage to equipment when not in use.

Store Equipment:

- In a cool, dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.) which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

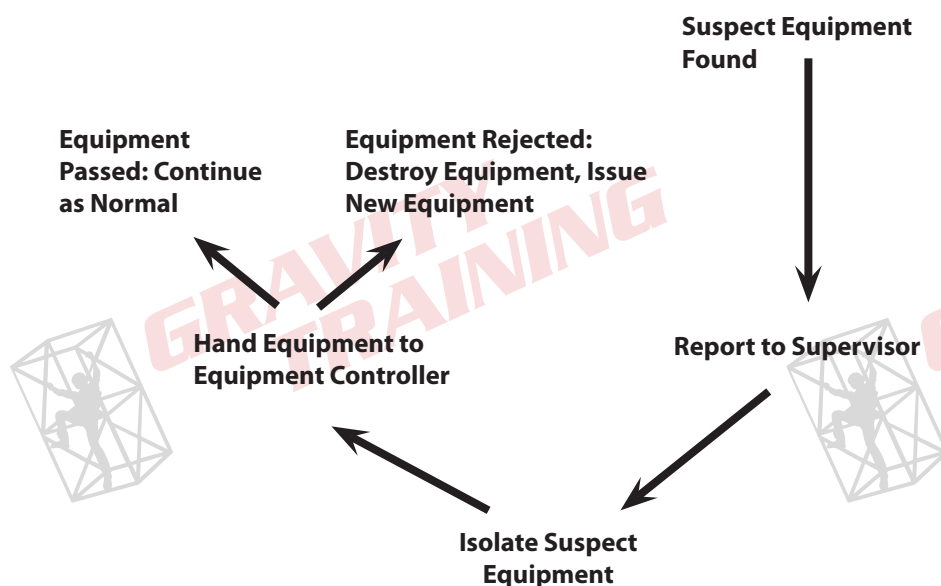


- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges, use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.

3.12 Reporting Suspect Equipment

Equipment that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or expired equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer.

NO CONTRACTOR MAY REPAIR PPE.

All the procedures pertaining to equipment will be included in the Fall Protection Plan. It is the duty of the Fall Protection Plan Implementer to ensure that these are being followed on site. This will include requesting inspection records, rejection notices, requisition forms etc. to ensure that the workers on site are complying to the relevant procedures.

10(2) (e) a rescue plan detailing the necessary procedure, personnel and suitable equipment required to affect a rescue of a person in the event of a fall incident to ensure that the rescue procedure is implemented immediately following the incident.

Before any work may start on site, one must consider the worst case scenarios which includes all emergency procedures as described in the HSE file, as well as the site-specific fall rescue plan.

It is the duty of the FPP Implementer to ensure that:

- All workers on site are aware of the rescue to be conducted.
- The rescue plan is applicable to the site.
- There must be a competent rescuer and a first aider on site.

The Fall Protection Plan will include a Rescue Plan, this plan will be site and task specific and will be written to the competencies of the rescuers on site. It is the duty of the Fall Protection Plan Implementer to ensure that the necessary equipment and personnel are present on site should a rescue need to be performed and that all workers are aware of the relevant emergency procedures. Proof that the workers are aware of these procedures and that all the necessary equipment and personnel are available on site, will be given in the form of induction training registers, pre-task risk assessments and rescue training certificates.

The rescue plan must be written by a competent Subject Matter Expert (SME) for task and access method.

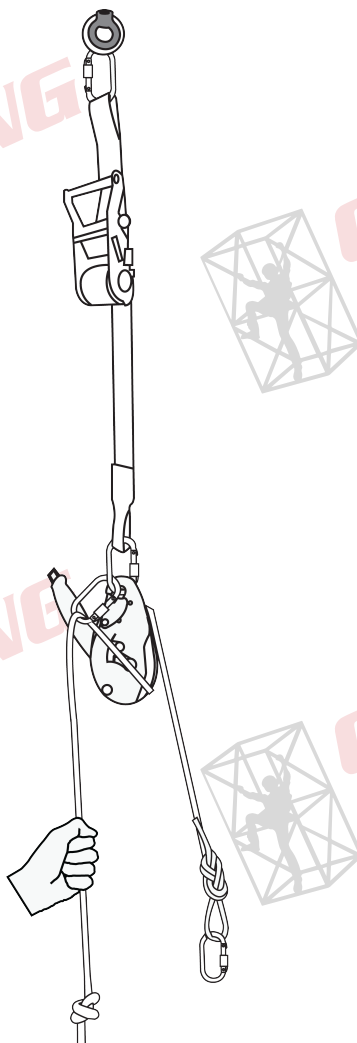
Ratchet Rescue

Rescue Procedure:

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to casualty's top D-link.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between the casualty and the descending device.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

When performing a rescue, remember that communication is of utmost importance and **MUST BE MAINTAINED AT ALL TIMES**.

To save time when attempting a rescue, as well as to reduce the risk of errors occurring and/or dropping items when at height, it is always a better to pre-rig the rescue system when on the ground before climbing up to the casualty.



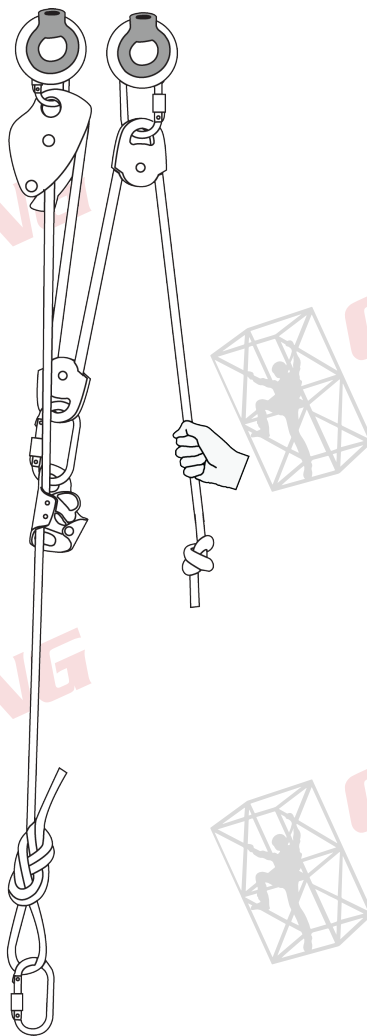
3:1 Rescue

Rescue Procedure

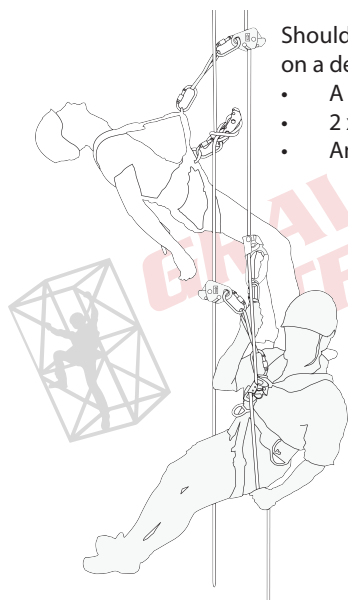
1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to casualty's top D-link.
4. Connect a suspension trauma sling.
5. Climb above the casualty.
6. Anchor the descending device to the structure.
7. Connect the Croll and pulley to rope.
8. Connect the rescue pulley to the structure.
9. Ensure all slack is removed between the casualty and the descending device.
10. Lift the casualty.
11. Remove the casualty's lanyard.
12. Remove the 3:1 system.
13. Apply extra friction.
14. Lower the casualty.

When performing a rescue, remember that communication is of utmost importance and **MUST BE MAINTAINED AT ALL TIMES.**

To save time when attempting a rescue, as well as to reduce the risk of errors occurring and/or dropping items when at height, it is always a better idea to rig the rescue system when on the ground before climbing up to the casualty.



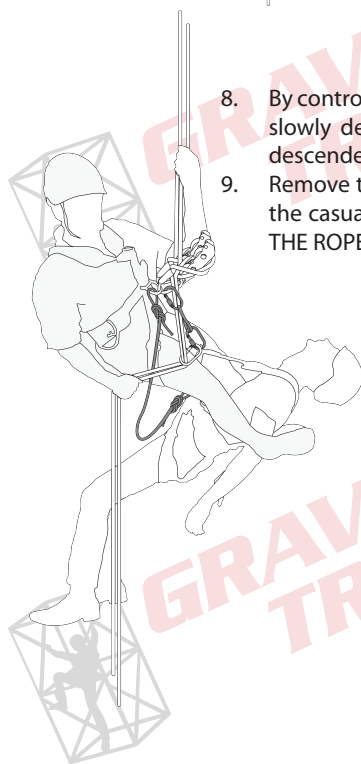
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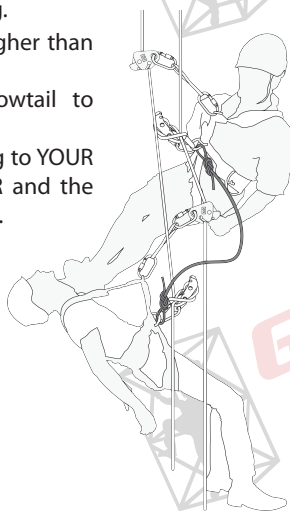
Should a worker, for any reason, be unable to reach the ground whilst on a descending device, a rescue must be performed. You will need:

- A rescue sling
- 2 x Karabiners
- An extra locker (to be connected to a cowtail):

1. Connect your backup device to the casualty's main line.
2. Ascend until you are the same level as the casualty.
3. Connect your cowtail locker above the casualty's stop.
4. Connect your locker sling to the locker above the casualty's stop and remove your cowtail.
5. Connect the casualty's cowtail to his/her locker and remove their locker sling.
6. Ascend until you are higher than the casualty.
7. Connect your extra cowtail to their BOTTOM D-LINK.
7. Connect the rescue sling to YOUR DESCENDER KARABINER and the CASUALTY'S TOP D-LINK.



8. By controlling the casualty's descender, slowly descend the casualty until the descender becomes slack.
9. Remove the casualty's descender from the casualty's harness, BUT NOT FROM THE ROPE.



10. Connect the casualty's descender to your harness; ENSURE THAT THE GATES OF BOTH KARABINERS ARE FACING DOWNWARD AND TOWARDS YOU, and remove all the slack and lock it.
11. Remove your backup device.
12. Use the karabiner connecting the casualty's locker sling to their harness and connect this karabiner to the slack ends of BOTH ropes.
14. Whilst keeping tension on the slack ends of the rope running through the extra karabiner, descend down in a controlled fashion.
15. Open both karabiners and remove your BOTTOM D-LINK.



3.14 CR - 10(1)(b)/(c) - Review & Monitoring Plan

Notes:

10(1)(b) ensure that the Fall Protection Plan contemplated in paragraph (a) is implemented, amended where and when necessary and maintained as required; and

10(1)(c) take steps to ensure continued adherence to the Fall Protection Plan.

A Fall Protection Plan must be reviewed to ensure that it is still up to date and that it still manages all fall risks on site.

The FPP Implementer need not review the plan him/herself, however it is the duty of the implementer to monitor the plan and inform the FPP Developer (CR (10(1)(a)) should the plan become insufficient or outdated, or if one of the following should occur:

- Should there be a change in legislation.
- Should there be a change in work scope.
- After an incident has occurred.
- After one year has passed.
- Should the client request it.

4. Ensuring the safety of work platforms

4.1 CR 10(4) A contractor must ensure that-

10(4) (a) all unprotected openings in floors, edges, slabs, hatchways and stairways are adequately guarded, fenced or barricaded or that similar means are used to safeguard any person from falling through such openings;

10(4) (b) no person is required to work in a fall risk position, unless such work is performed safely as contemplated in sub-regulation (2);

10(4) (c) fall prevention and fall arrest equipment are-

- (i) approved as suitable and of sufficient strength for the purpose for which they are being used, having regard to the work being carried out and the load, including any person, they are intended to bear; and
- (ii) securely attached to a structure or plant, and the structure or plant and the means of attachment thereto are suitable and of sufficient strength and stability for the purpose of safely supporting the equipment and any person who could fall; and

10(4) (d) fall arrest equipment is used only where it is not reasonably practicable to use fall prevention equipment.

4.2 CR 10(5) (e) - suitable and sufficient platforms, coverings or other similar means of support have been provided to be used in such a way that the weight of any person passing across or working on or from fragile material is supported; and (t) suitable and sufficient guard-rails, barriers and toe-boards or other similar means of protection prevent, as far as is reasonably practicable, the fall of any person, material or equipment.

4.3 CR 11(2) An owner of a structure must ensure that-

11(2) (a) inspections of that structure are carried out periodically by competent persons in order to render the structure safe for continued use;

11(2) (b) that the inspections contemplated in paragraph (a) are carried out at least once every six months for the first two years and thereafter yearly;

11(2) (c) the structure is maintained in such a manner that it remains safe for continued use;

11(2) (d) the records of inspections and maintenance are kept and made available on request to an inspector.

Notes:

4.4 CR 16 Scaffolding (SANS 10085).

(1) A contractor must appoint a competent person in writing who must ensure that all scaffolding work operations are carried out under his or her supervision and that all scaffold erectors, team leaders and inspectors are competent to carry out their work.

(2) A contractor using access scaffolding must ensure that such scaffolding, when in use, complies with the safety standards incorporated for this purpose into these Regulations under section 44 of the Act.

4.5 GSR 13A - Ladders (EN 131)

1. An employer shall ensure that every ladder is constructed of sound material and is suitable for the purpose for which it is used, and -
 - a) is fitted with non-skid devices at the bottom ends and hooks or similar devices at the upper ends of the stiles to ensure the stability of the ladder during normal use; or;
 - b) is so lashed, held or secured whilst being used as to ensure the stability of the ladder under all conditions and at all times.
2. No employer shall use a ladder, or permit it to be used, if it -
 - a) i) has rungs fastened to the stiles only by means of nails, screws, spikes or in like manner; or
 - ii) has rungs which have not been properly let into the stiles: Provided that, in the case of welded ladder or ladders of which the rungs are bolted or riveted to the stiles, the rungs need not be let into the sides; or
 - b) has damaged stiles, or damaged or missing rungs.

4.6 CR 17 - Suspended platforms

17 (1) A contractor must appoint a competent person in writing who must ensure that all suspended platforms work operations are carried out under his or her supervision and that all suspended platform erectors, operators and inspectors are competent to carry out their work.

ALL STRUCTURES/PLATFORMS/LADDERS MUST COMPLY TO THE RELEVANT SANS OR EN STANDARDS. THE DEVELOPER MUST FAMILIARISE HIM/HERSELF WITH THE REQUIRED STANDARDS BEFORE WRITING A FALL PROTECTION PLAN.

The Fall Protection Plan Implementer will not be responsible for the certification, testing or inspection of any work platform. The Fall Protection Plan will state that the structures will be safe to work on as per the work at height risk assessment. The Implementer must ensure that he/she obtains proof that this is indeed the case, either in the form of the baseline risk assessment, provided by the client as per CR 5, hand-over certificates, inspection registers, scaff-tags, operating certificates or engineer's permission that the site is indeed safe.

18(1) A contractor must-

18(1)(a) appoint a competent person in writing as a rope access supervisor with the duty of supervising all rope access work on the site, including the duty of ensuring occupational health and safety compliance in relation to rope access work, provided that the appointment of any such person does not relieve the construction manager of any personal accountability for failing in his management duties in terms of this regulation;

18(1) (b) ensure that all rope access work on the construction site is carried out under the supervision of a competent person; and

18(1) (c) ensure that all rope access operators are competent and licensed to carry out their work.

18(2) No contractor may use or allow the use of rope access work unless-

18(2)(a) the design, selection and use of the equipment and anchors comply with the safety standards incorporated for this purpose into these Regulations under section 44 of the Act; and

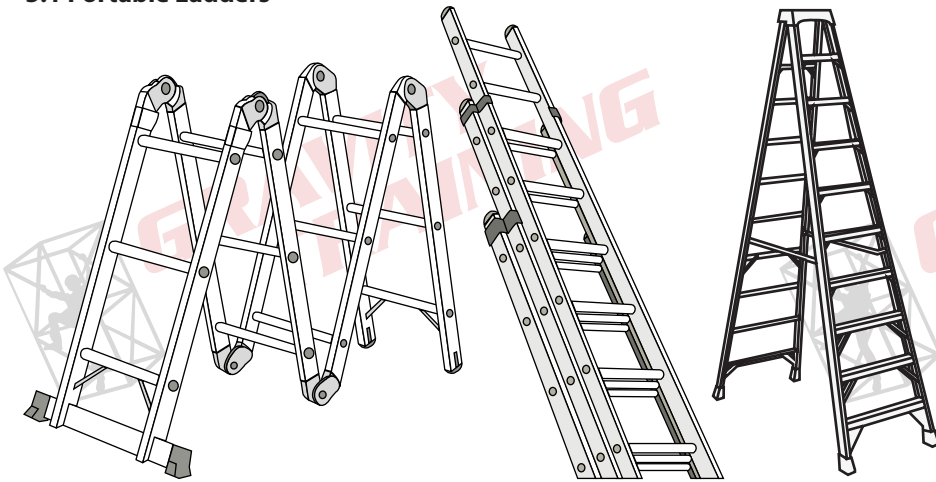
18(2) (b) he or she is in possession of a site specific Fall Protection Plan developed by a competent person applicable to the specific work and environment prior to the commencement of the work, including records of maintenance and inspections of all the equipment used for the work operations.

18 (3) A contractor must ensure that adequate measures are in place to allow rescue procedures to commence immediately in the event of a fall incident taking place.

4.8 Section 44 - Incorporation of health and safety standards in regulations

Sec44(1) The Minister may by notice in the Gazette incorporate in the regulations any health and safety standard or part thereof, without stating the text thereof, by mere reference to the number, title and year of issue of that health and safety standard or to any other particulars by which that health and safety standard is sufficiently identified.

5.1 Portable Ladders

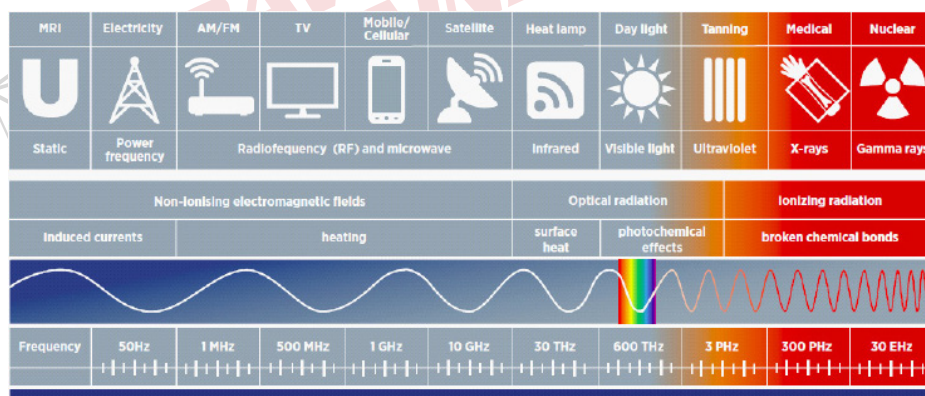


Portable ladders account for the majority of falls from height due to the general assumption that people know how to use them correctly. Falls from ladders are generally caused by:

- Wrong type of ladder being used for the task.
- Ladder is not safe for use.
- Ladder has not been manufactured to the relevant standard.
- Workers have not been trained on how to use ladders.
- Ladders transported incorrectly.

It should be clear that portable ladders add significant risk to a work site, and should thus be mitigated by ensuring that all work carried out on portable ladders is done so by trained and competent technicians. Ensure that this is included in the training matrix.

5.2 RF Radiation



Due to the increase in use of electronic devices, the amount of RF radiation has been increasing significantly, resulting in most open spaces being used to erect antennas and/or dishes for the broadcasting and/or receiving of RF signals. The radiation given off by such devices pose a significant risk to workers as a result of:

- Prolonged exposure that can cause serious internal and external injury.
- Workers may suffer electrical shock/burns due to unfamiliarity with equipment and/or procedures.
- Eye injury.

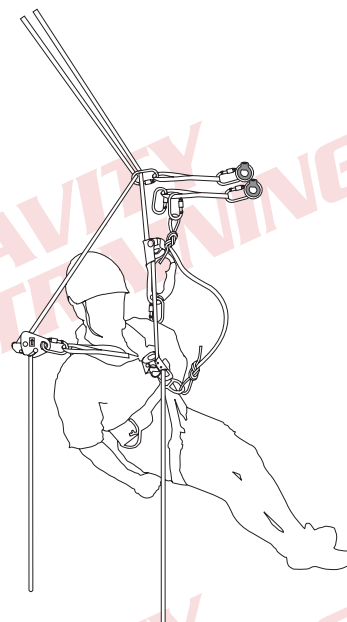
It should be clear that RF radiation add significant risks to a work site, and should thus be mitigated by ensuring that all work carried out in RF areas is done so by trained and competent technicians. Ensure that this is included in the training matrix.

5.3 Rope Access

On many work sites, contractors employ Rope Access to gain access to areas where other means would either be impossible or substantially more expensive. Although Rope Access is a safe system, there are certain considerations that must be kept in mind when dealing with Rope Access on site.

- Rope Access must be conducted (in South Africa) in accordance with CR 18.
- Rope Access must be conducted under the rules and regulations set forth by the relevant governing body (IWH, IRATA, etc.). These bodies stipulate which standards are to be followed as well as best practice notes which must be followed on site. Ensure that these are included in the risk assessment.
- Rope Access work must be conducted under competent supervision.
- Only competent technicians (refer to training matrix for the relevant training requirements) are allowed to perform Rope Access work.
- Rope Access must be conducted in at least teams of two.
- Even if the Rope Access technicians are on another contractor's team, this still poses a risk as they may be working above other technicians that poses an additional risk which needs to be mitigated.

As mentioned above, Rope Access is a safe system, used world wide, but can pose fatal risks should it be attempted by incompetent persons. Ensure that the risk assessment and training matrix stipulate sufficient mitigations.

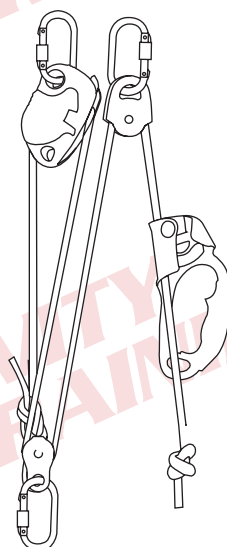


5.4 Rope Rigging (1-100kg)

On many work sites, contractors may need to use Rope Rigging to lift, lower or move objects utilising rope and pulley systems. This subjects not only the Rope Rigging Technicians to risk, but also the other people on site.

- Rope Rigging must be conducted in accordance with a **Lifting Plan**, written by a competent person.
- Rope Rigging must be conducted by competent persons.
- Rope Rigging must be conducted in teams of at least 2 (depending on the type and number of system/s to be used).
- Barricading and drop-zones are to be used as needed.

As mentioned above, Rope Rigging can pose fatal risks should it be attempted by incompetent persons. Ensure that the risk assessment and training matrix stipulate sufficient mitigations.

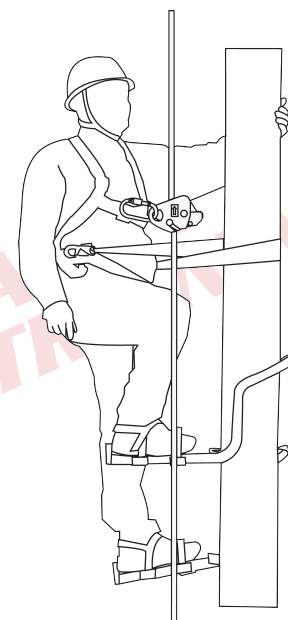


5.5 Alternative/Unfamiliar Access Methods

On some sites, it may transpire, that workers are presented with an Access Method they have not previously seen or one which they have not been specifically trained on. This creates an especially dangerous environment as workers may attempt to perform a task with no prior experience and/or knowledge of how to do so safely.

- Technicians are to be trained on all relevant Access Methods before starting work.
- Supervisor to ensure all Access Methods are made safe as per the relevant principles (Fall Arrest, Rope Access, Work Restraint etc.).

All requirements for any specialised Access Methods should be made clear within the risk assessment as well as the training matrix.



6. Policies

Now that the foundation has been laid with regards to the relevant requirements that must be included in a Fall Protection Plan, we can establish how to convert these regulations and standards into a usable plan.

A company must provide proof of their commitment to conduct work in accordance with the relevant requirements. To do this, a policy containing the following, should be written:

A reason - WHY the policy exists and WHY it should be adhered to.

A responsible person - The person WHO will be charged with ensuring that the policy is implemented. It is recommended that the person is not named, but rather referred to by designation. This will prevent having to rewrite the plan whenever there is a change of personnel.

Task - Put quite simply, WHAT must be done by the responsible person. Ensure that no task contradicts any of the regulations or standards mentioned previously. Refer to the relevant regulations where applicable.

Time - It must be stipulated WHEN a given task will be done by a responsible person. Typically, most policies must be implemented "before work starts". Again, be careful not to be too specific with these stipulations, as this may lead to liability issues in the case of an incident or accident.

Proof - All policies must be implemented and as such proof of said implementation must be available. For example: signed registers of induction, valid COF's etc.

An example of a risk assessment policy:

"In order to create and maintain a safe work environment a Work at Height Risk Assessment must be written and conducted by a competent risk assessor as per CR 9 before work starts."

In the example above:

- The WHY is to create and maintain a safe work environment.
- The WHO is the competent risk assessor.
- The WHAT is the Work at Height Risk Assessment.
- The WHEN is before work starts.
- The PROOF will be the physical risk assessment.
- All this must be done in accordance with CR 9.

Notes:

7. Standard Operating Procedures

The standard operating procedures specify the exact steps that need to be taken in order to perform a given task. It is recommended that the personnel involved in performing the tasks are included in the development process.

These SOP's are task and site specific. Sometimes it may be possible to write one SOP that can be applied to various sites, for example: using fall arrest equipment will always be done as per training matrix. However, this might not always be the case, thus ensure that all SOP's are applicable to the relevant site.

SOP's must include the following:

- Scope and purpose - Describe the reason for the existence of the SOP and the affected personnel/departments.
- Equipment/supplies/materials - What external items are required to complete a given task.
- Methodology - List all the steps with the with necessary detail in order to complete a given task.
- "What-if's" and Interferences - What can go wrong and what are the steps to be taken should something outside of the SOP occur.

Review Process - SOP's need to be reviewed after an incident or accident has occurred and/or in accordance with the company's normal review schedule. It is also recommended that the persons involved in carrying out the SOP are included in the review process.

Notes:

8. Incident Reporting

24. Report to inspector regarding certain incidents

- (1) Each incident occurring at work or arising out of or in connection with the activities of persons at work, or in connection with the use of plant or machinery, in which, or in consequence of which-
- (a) any person dies, becomes unconscious, suffers the loss of a limb or part of a limb or is otherwise injured or becomes ill to such a degree that he is likely either to die or to suffer a permanent physical defect or likely to be unable for a period of at least 14 days either to work or to continue with the activity for which he was employed or is usually employed;

Any and all incidents must be reported in accordance with Section 24 of the OHS Act, using the relevant document templates provided (GAR Annexure 1).

Should there be any operation that falls outside the relevant accepted standards (e.g. operations outside the OEM requirements) work must be stopped and the transgression reported to the Construction Manager/Risk Assessor/Safety Officer/HSE Representative.

Any and all reported transgressions must be documented and recorded with as much detail as possible. This will aid the investigation, should there be one, as well as allow the cause of such transgressions to be identified and removed. Workers must report any and all non-conformances in order for an investigation to be conducted so that a safe work site may be established and maintained.

Should the Fall Protection Implementer be unsure of the course of action he/she must contact the Fall Protection Plan Developer in order to determine the best way forward. All incidents must be recorded and investigated in order to determine the source/cause of the incident so that it may be prevented in future helping to maintain an as-safe-as-possible work site

9. Example Site

Write a Fall Protection Plan based on the site given below as a class exercise using the Fall Protection Plan Sample provided. All the documents within the Fall Protection Plan sample is provided on the Gravity Training 229994 CD.

The task is as follows:

- The fibre optic cable that leads to the booster antenna on the top of the pitched roof needs replacing.
- There are 3 workers on site.
- You can use any access method you wish.
- The height at the base of the antenna is 8m.
- The building is surrounded by grass at the base.
- The building has bricked walls.
- It is a tiled, pitched roof.
- It is a bright, sunny day.
- The technicians will use basic hand tools to replace the cable.

You may work in groups.

